

Maths Class: _____

2009

Time Allowed: 3 hours plus 5 mins reading time

- Write your name at the top of this page, and at the top of each answer sheet
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new answer page.

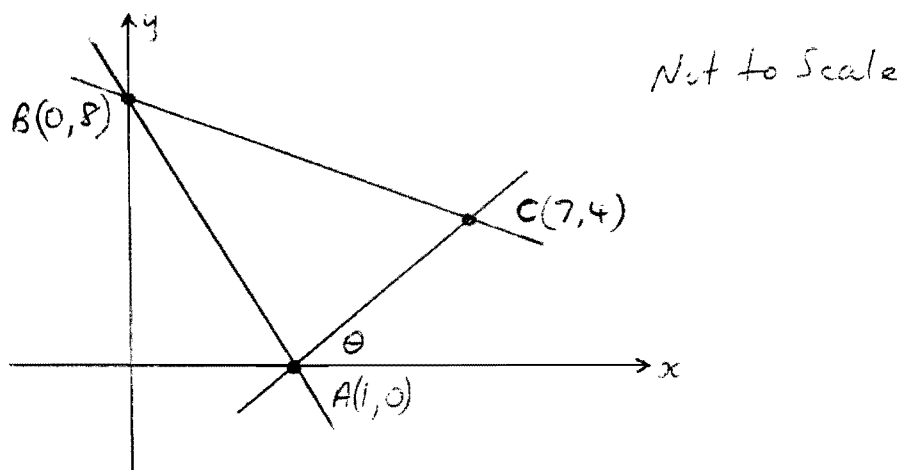
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Question 1

- a) Calculate $(9.6 \times 10^4) \div (6.3 \times 10^{-2})$. Write your answer using 3 significant figures. 1
- b) Solve i) $x^2 - 3x - 4 = 0$ 2
 ii) $x^3 + x^2 + x + 1 = 0$ 2
- c) i) Sketch $y = |x + 1|$. Show intercepts on the coordinate axes. 1
 ii) Hence or otherwise solve $|x + 1| = x + 1$ 1
- d) Find the gradient of the curve $y = \frac{1}{x^2}$ at the point $(-1, 1)$ 2
- e) Simplify $\frac{x+4}{x} + \frac{3}{x^2-x}$ 2
- f) Write the exact value of $\tan \frac{5\pi}{6}$ 1

Question 2 (start a new page)

a)



Points A, B, C have coordinates as shown above.

The angle between AC and the x axis is θ .

- i) Copy this diagram into your answer booklet.
- ii) Find the gradient of AC .

1

- iii) Calculate the size of θ to the nearest degree. 1
- iv) Find the equation of AC . Give your answer in general form. 2
- v) Find the equation of the perpendicular bisector of the interval AC . 3
- vi) Find the perpendicular distance from B to AC . 2
- vii) Write the coordinates of D such that $ABCD$ is a parallelogram. 1
- b) Solve $\cos\theta = -\frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$ 2

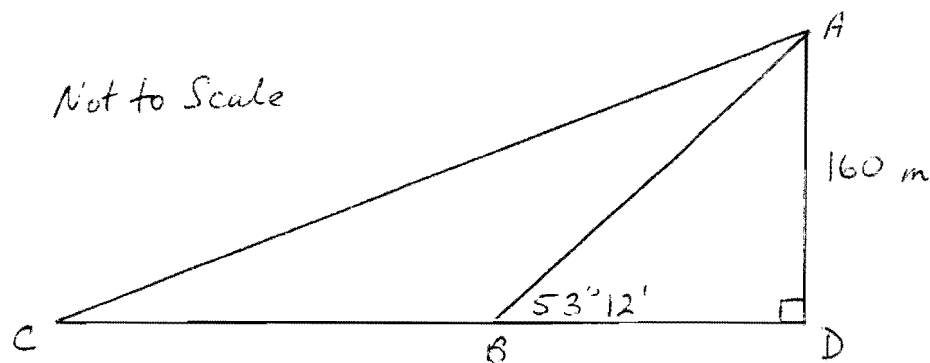
Question 3 (start a new page)

- a) Differentiate:
 - i) $y = \frac{2x+1}{2x-1}$ 2
 - ii) $y = (x^2 - 1)^4$ 1
 - iii) $y = 3x \log x$ 2
 - iv) $y = \sin^2 x$ 1
 - v) $y = \tan 2x$ 1
- b) Find indefinite integrals of:
 - i) $\cos 5x$ 1
 - ii) $\frac{2x}{x^2+1}$ 1
 - iii) $(2x + 3)^5$ 1
- c) Given $f(x) = x^2$, show how to find $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ 2

Question 4 (start a new page)

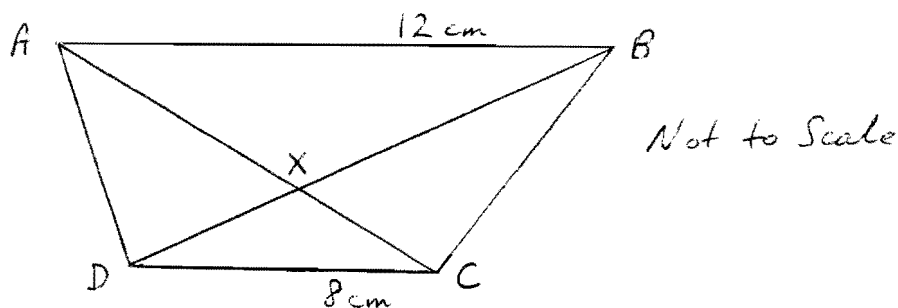
- a) A parabola has equation $2y = x^2 - 6x + 7$
- Write this equation in the form $(x - h)^2 = 4a(y - k)$ and hence, or otherwise, give the coordinates of the vertex. 2
 - Find the coordinates of the focus. 1
 - Give the equation of the directrix. 1

b)



A man in a boat at B observes that the angle of elevation of a cliff top 160 metres high is $53^\circ 12'$.

- Find the direct distance BA. Give your answer correct to 1 decimal place. 2
 - The man rows 50 metres from B to C. Use the cosine rule in triangle ABC to find AC to the nearest metre. 2
- c) ABCD is a trapezium in which $AB \parallel DC$, $AB = 12\text{ cm}$, $DC = 8\text{ cm}$ and $AC = 9\text{ cm}$. The diagonals intersect at X.



- Copy the diagram onto your answer page and clearly mark the above information.
- Prove that $\triangle AXB$ is similar to $\triangle CXD$. 2
- Hence find the length of AX. 2

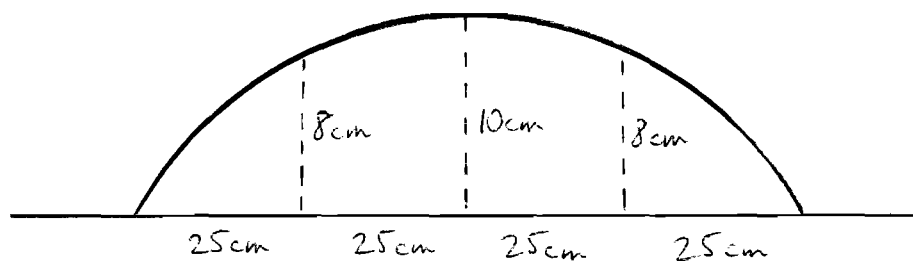
Question 5 (start a new page)

- a) Find the value(s) of k so that $x^2 - (k + 2)x + 4k - 4 = 0$ has:
- i) equal roots. 2
 - ii) one root the reciprocal of the other. 2
- b) Simplify $\frac{\sin(\frac{\pi}{2} - \theta)}{\sin(\pi - \theta)}$ 2
- c) Solve $\sin^2 x + \sin x = 0$ for $0 \leq x \leq 2\pi$ 2
- d) i) Sketch the graphs of $y = \frac{1}{x}$, $y = 1$ and $x = 2$ on the same axes. 2
Shade the common area between the three graphs.
- ii) Find the value of the shaded area in exact form. 2

Question 6 (start a new page)

- a) Solve $3^x = 7$. Give your answer correct to 2 decimal places. 2
- b) The first three terms of a certain geometric series are $x + 2 + 1^{\frac{1}{2}} + \dots$
- i) Find the value of x . 1
 - ii) Find the limiting sum of the series. 1
- c) Evaluate $\sum_{n=10}^{50} (100 - 4n)$ 2

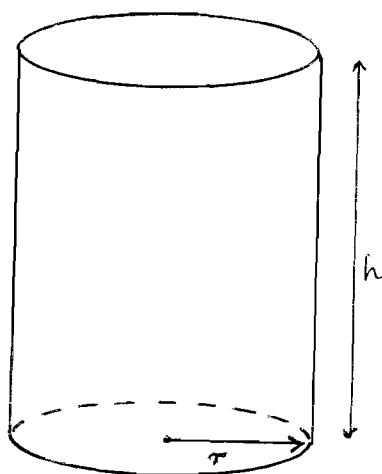
- d) A speed hump has a cross-section as shown:



Use Simpson's rule to find the approximate area of the cross-section.

2

- e)



The cylinder varies its volume but the sum of its radius and height is constant at 12 cm.

- i) Given $V = \pi r^2 h$, find $\frac{dv}{dr}$.

2

- ii) Find the radius that proves the volume is maximised.

2

Question 7 (start a new page)

- a) The curve $y = e^{x+1}$ is rotated about the x axis between $x = 0$ and $x = 1$.
Find the volume generated, leaving your answer in exact form.

3

- b) Given the function $y = -2\sin 3x$.

- i) State the period of the function.

1

- ii) Sketch its graph for $0 \leq x \leq \pi$

2

c) For a function $y = f(x)$, you are given that $\frac{dy}{dx} = 3x^2 - 2x - 1$.

There is a stationary point at (1,2).

i) Determine the nature of the stationary point. 2

ii) Prove that there is a point of inflexion. 2

iii) Find the equation of the function $y = f(x)$. 2

Question 8 (start a new page)

a) i) Solve $4x - x^2 \geq 0$ 2

ii) Hence, state the range of the function $y = \frac{-1}{\sqrt{4x-x^2}}$ 1

b) i) Prove that $1 + \tan^2 x = \sec^2 x$ 2

ii) Hence, find $\int \tan^2 3x \, dx$ 2

c) Solve $\log_2 x + \log_2(x^3) = -8$ 2

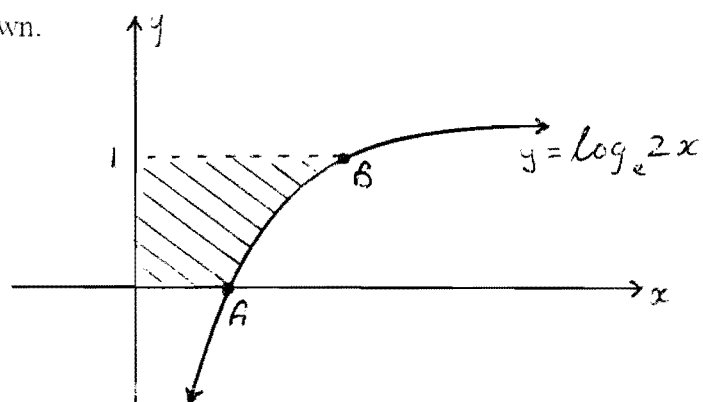
d) The sum of the first two terms of a geometric sequence is 6 and the sum of the second and third terms is -5. 3

Find the first three terms of the sequence.

Question 9 (start a new page)

a) Find $\int \frac{x^3+1}{2x} \, dx$ 2

- b) The graph of $y = \log_e 2x$ is shown.



- i) Find the coordinates of A and B. 2
 - ii) Find the shaded area. 3
- c) \$1000 is deposited into a savings account.
- i) If the account earns 6% pa compounded annually, find the account's value at the end of 15 years. 1
 - ii) If the account is to have a value of \$5000 after 15 years, find the annual compound interest rate needed to achieve this.
Give your answer to the nearest whole percent. 2
 - iii) \$1000 is deposited at the beginning of each year into the account paying 6% pa. What will be the total value of the account at the end of 15 years, to the nearest dollar. 2

Question 10 (start a new page)

- a) A block of ice, with mass M grams after t minutes is melting according to the equation $M = Mo e^{-kt}$.
- Initially, there is 100 grams of ice and, after 35 minutes, the block has reduced to 60 grams.
- i) Find the constants Mo and k (exact form). 3
 - ii) Find the rate of melting when the ice weighs 50 grams. 1
 - iii) Find how long it takes for the ice to reduce to 5 grams. 2

- b) A particle moves so that its position x *cm* from a fixed point 0 after t seconds, is given by $x = 2te^{-t}$.
- i) Find the particle's initial position. 1
- ii) Show that $v = 2e^{-t}(1 - t)$. 1
- iii) Find when the particle is stationary. 1
- iv) Describe the particle's position as $t \rightarrow \infty$ 1
- v) Use the above information to sketch, on coordinate axes, a graph of the particle's position over time.
 Show on your curve the time where you expect the acceleration to be zero.
 Justify the location. 2

END OF PAPER.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONS

a) 1520000

b) i) $(x-4)(x+1) = 0$

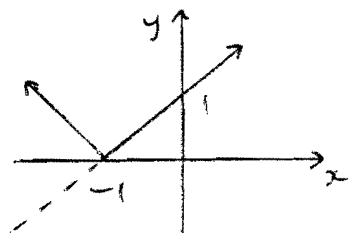
$x = 4, -1$

ii) $x^2(x+1) + 1(x+1) = 0$

$(x+1)(x^2+1) = 0$

$x = -1$

c) i)



ii) $x \geq -1$

d) $y = x^{-2}, \frac{dy}{dx} = -2x^{-3}$

$m_T = \frac{-2}{(-1)^3}$
 $= 2$

e) $\frac{(x+4)(x-1)}{x(x-1)} + \frac{3}{x(x-1)}$

$= \frac{x^2 + 3x - 4 + 3}{x(x-1)}$

$= \frac{x^2 + 3x - 1}{x(x-1)}$

f) $-\frac{1}{\sqrt{3}}$

iv) $y - 0 = \frac{2}{3}(x - 1)$

$3y = 2x - 2$

$2x - 3y - 2 = 0$

v) $m.p. = (4, 2), m_2 = -\frac{3}{2}$

$\therefore y - 2 = -\frac{3}{2}(x - 4)$

$2y - 4 = -3x + 12$

$3x + 2y - 16 = 0$ ($y = -\frac{3x}{2} + 8$)

vi) $p.d. = \left| \frac{2 \times 0 - 3 \times 8 - 2}{\sqrt{2^2 + (-3)^2}} \right|$

$= \left| \frac{-26}{\sqrt{13}} \right|$

$= \frac{26}{\sqrt{13}}$ (or $2\sqrt{13}$)

vii) $D(8, -4)$

b) $\theta = 60^\circ$ (2nd, 3rd quadrants.)

$= 120^\circ, 240^\circ$

② i) diagram

ii) $M_{AC} = \frac{4-0}{7-1}$
 $= \frac{2}{3}$

iii) $\tan \theta = \frac{2}{3}$
 $\therefore \theta \doteq 34^\circ$

$$\begin{aligned} \textcircled{3} \text{ a) i) } \frac{dy}{dx} &= \frac{2(2x-1) - 2(2x+1)}{(2x-1)^2} \\ &= \frac{4x-2-4x-2}{(2x-1)^2} \\ &= \frac{-4}{(2x-1)^2} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{dy}{dx} &= 4(x^2-1)^3 \times 2x \\ &= \underline{\underline{8x(x^2-1)^3}} \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{dy}{dx} &= 3 \log x + \frac{1}{x} \cdot 3x \\ &= \underline{\underline{3 \log x + 3}} \end{aligned}$$

$$\text{iv) } \frac{dy}{dx} = \underline{\underline{2 \sin x \cos x}}$$

$$\text{v) } \frac{dy}{dx} = \underline{\underline{2 \sec^2 2x}}$$

$$\text{b) i) } \frac{\sin 5x}{5} + c$$

$$\text{ii) } \log(x^2+1) + c$$

$$\text{iii) } \frac{(2x+3)^6}{12} + c$$

$$\begin{aligned} \text{c) } \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} \\ &= \underline{\underline{2x}} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \text{ a) i) } x^2 - 6x &= 2y - 7 \\ x^2 - 6x + 9 &= 2y + 2 \\ \underline{\underline{(x-3)^2}} &= \underline{\underline{2(y+1)}} \end{aligned}$$

$\therefore$ vertex is (3, -1)

$$\text{ii) } 4a = 2 \Rightarrow a = \frac{1}{2}$$

$\therefore$ Focus is (3, -\frac{1}{2})

$$\text{iii) } \underline{\underline{y = -\frac{1}{2}}}$$

$$\text{b) i) } \sin 53^\circ 12' = \frac{160}{BA}$$

$$\begin{aligned} \therefore BA &= \frac{160}{\sin 53^\circ 12'} \\ &= \underline{\underline{199.8 \text{ m}}} \end{aligned}$$

$$\begin{aligned} \text{ii) } AC^2 &= 50^2 + 199.8^2 - 2 \times 50 \times 199.8 \\ &\quad \times \cos 126^\circ 48' \\ &= 54388.5315 \end{aligned}$$

$$\therefore AC \doteq \underline{\underline{233 \text{ metres}}}$$

c) i) diagram

ii) $\angle A = \angle C$ (alternate angles
AB || DC)

$\angle B = \angle D$ (same reason)

$\therefore \triangle AXB \parallel \triangle CXD$ (equiangular)

$$\text{iii) } \frac{AX}{CX} = \frac{12}{8} = \frac{3}{2} \text{ (equal ratio of sides in sim. triangles)}$$

$$\begin{aligned} \therefore AX &= \frac{3}{5} \times 9 \\ &= \underline{\underline{5.4 \text{ cm}}} \end{aligned}$$

5) a) i) $b^2 - 4ac = 0$

$$[-(k+2)]^2 - 4 \times 1 \times (4k-4) = 0$$

$$k^2 + 4k + 4 - 16k + 16 = 0$$

$$k^2 - 12k + 20 = 0$$

$$(k-10)(k-2) = 0$$

$$\therefore \underline{k = 10, 2}$$

ii) $\alpha = \frac{1}{\beta}$

$$\therefore \alpha\beta = 1$$

$$\therefore \frac{c}{a} = 1$$

$$\therefore \frac{4k-4}{1} = 1$$

$$\therefore 4k = 5$$

$$\underline{k = \frac{5}{4}}$$

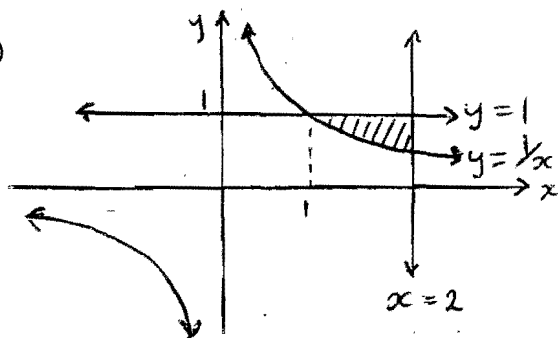
b) $\frac{\cos \theta}{\sin \theta} = \underline{\cot \theta}$

c) $\sin x (\sin x + 1) = 0$

$$\sin x = 0 \text{ or } -1$$

$$\therefore \underline{x = 0, \pi, 2\pi, \frac{3\pi}{2}}$$

d) i)



ii) $\text{Area} = 1 - \int_1^2 \frac{1}{x} dx$

$$= 1 - [\log]_1^2$$

$$= 1 - \log 2 + \log 1$$

$$= 1 - \log 2$$

$$= \sqrt{-1 + e^{-2}} + e^2 - 2 + 1$$

$$= \sqrt{1 + e^{-2}} + e^2 - 3$$

$$= \underline{\underline{e^{-2} + e^2 - 3}}$$

6) a) $\log(3^x) = \log 7$

$$x \log 3 = \log 7$$

$$x = \frac{\log 7}{\log 3}$$

$$\doteq \underline{\underline{1.77}}$$

b) i) $\frac{2}{x} = \frac{1\frac{1}{2}}{2} \left(\frac{3}{4}\right)$

$$3x = 8$$

$$\underline{x = 2\frac{2}{3}}$$

ii) $a = 2\frac{2}{3}, r = \frac{3}{4}$

$$S_{\infty} = \frac{2\frac{2}{3}}{1 - \frac{3}{4}}$$

$$= 2\frac{2}{3} \times 4$$

$$= \underline{\underline{10\frac{2}{3}}}$$

c) $a = 60, d = -100, n = 41$

$$S_{41} = \frac{41}{2} (60 - 100)$$

$$= \underline{\underline{-820}}$$

d) $\text{Area} \doteq \frac{25}{3} (0 + 4 \times 8 + 10) \times 2$

$$= \underline{\underline{700 \text{ cm}^2}}$$

e) i) $r + h = 12 \Rightarrow h = 12 - r$

$$V = \pi r^2 h$$

$$= \pi r^2 (12 - r)$$

$$= 12\pi r^2 - \pi r^3$$

$$\frac{dV}{dr} = \underline{\underline{24\pi r - 3\pi r^2}}$$

⇒ ii) Max volume when $\frac{dV}{dr} = 0$

$$\therefore 24\pi r - 3\pi r^2 = 0$$

$$3\pi r(8-r) = 0$$

$$\therefore r = 0, 8$$

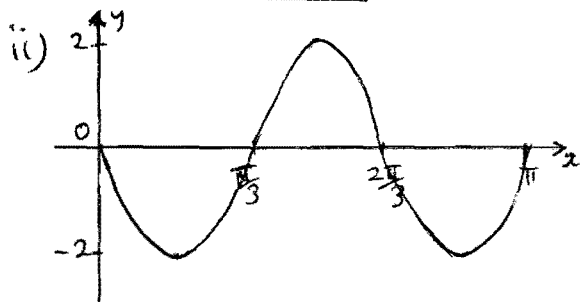
$$\frac{d^2V}{dr^2} = 24\pi - 6\pi r$$

$$\text{When } r = 8, \frac{d^2V}{dr^2} < 0$$

∴ Volume is maximised
when radius = 8 cm.

⑦ a) $V = \pi \int_0^1 (e^{x+1})^2 dx$
 $= \pi \int_0^1 e^{2x+2} dx$
 $= \pi \left[\frac{1}{2} e^{2x+2} \right]_0^1$
 $= \pi \left(\frac{1}{2} e^4 - \frac{1}{2} e^2 \right)$
 $= \underline{\underline{\frac{\pi}{2} (e^4 - e^2) u^3}}$

b) i) period = $\underline{\underline{\frac{2\pi}{3}}}$



c) i) $\frac{d^2y}{dx^2} = 6x - 2$

$$\text{When } x = 1, \frac{d^2y}{dx^2} > 0$$

∴ maximum turning
point at (1, 2)

i) Point of inflexion when $\frac{d^2y}{dx^2} = 0$
and changes concavity.

$$\therefore 6x - 2 = 0 \Rightarrow \text{test } x = \frac{1}{3}$$

x	$\frac{1}{3}^-$	$\frac{1}{3}$	$\frac{1}{3}^+$
$\frac{d^2y}{dx^2}$	-	0	+

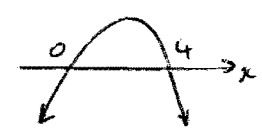
∴ concavity changes
∴ a pt. of inflexion
occurs when $x = \frac{1}{3}$.

ii) $y = \int (3x^2 - 2x - 1) dx$
 $= x^3 - x^2 - x + c$

Sub $x = 1, y = 2$

$$\therefore 2 = 1 - 1 - 1 + c \quad (c = 3)$$

$$\therefore \underline{\underline{y = x^3 - x^2 - x + 3}}$$

⑧ a) i) $x(4-x) \geq 0$ 
 $\therefore \underline{\underline{0 \leq x \leq 4}}$

ii) $\underline{\underline{y \leq -\frac{1}{2}}}$

b)

i) LHS = $1 + \tan^2 x$

$$= 1 + \frac{\sin^2 x}{\cos^2 x}$$

$$= \frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$

$$= \text{RHS}$$

$$\begin{aligned} \text{ii) } \int \tan^2 3x dx &= \int (\sec^2 3x - 1) dx \\ &= \frac{\tan 3x}{3} - x + c \\ &= \underline{\underline{\frac{\tan 3x}{3} - x + c}} \end{aligned}$$

$$\begin{aligned} \text{c) } \log_2 x + 3 \log_2 x &= -8 \\ 4 \log_2 x &= -8 \\ \log_2 x &= -2 \\ \therefore x &= 2^{-2} \\ &= \underline{\underline{\frac{1}{4}}} \end{aligned}$$

$$\begin{aligned} \text{d) } a + ar &= 6, ar + ar^2 = -5 \\ a(1+r) &= 6 \quad \text{--- ①} \\ ar(1+r) &= -5 \quad \text{--- ②} \\ \text{②} \div \text{①} : r &= -\frac{5}{6} \\ \text{Sub in ①} : a &= \frac{6}{\frac{1}{6}} \\ &= 36 \end{aligned}$$

$$\therefore \underline{\underline{36, -30, 25}}$$

$$\begin{aligned} \text{⑨ a) } \int \left(\frac{x^3}{2x} + \frac{1}{2x} \right) dx \\ &= \int \left(\frac{x^2}{2} + \frac{1}{2x} \right) dx \\ &= \underline{\underline{\frac{x^3}{6} + \frac{1}{2} \log x + c}} \end{aligned}$$

$$\text{b) i) } \underline{\underline{A\left(\frac{1}{2}, 0\right), B\left(\frac{e}{2}, 1\right)}}$$

$$\text{ii) } e^y = 2x \Rightarrow x = \frac{1}{2} e^y$$

$$\begin{aligned} \text{Area} &= \int_0^1 \frac{1}{2} e^y dy \\ &= \left[\frac{1}{2} e^y \right]_0^1 = \underline{\underline{\left(\frac{1}{2} e - \frac{1}{2} \right) u^2}} \end{aligned}$$

$$\begin{aligned} \text{c) i) } A &= 1000(1.06)^{15} \\ &= \underline{\underline{\$2396.56}} \end{aligned}$$

$$\begin{aligned} \text{ii) } 5000 &= 1000(1+r)^{15} \\ 5 &= (1+r)^{15} \\ \therefore r &= \sqrt[15]{5} - 1 \\ &\doteq 0.11326 \\ &\doteq \underline{\underline{11\%}} \end{aligned}$$

$$\begin{aligned} \text{ii) First } \$1000 &\Rightarrow 1000(1.06)^{15} \\ \text{Second } \$1000 &\Rightarrow 1000(1.06)^{14} \\ &\vdots \\ \text{Last } \$1000 &\Rightarrow 1000(1.06)^1 \\ \text{Total} &= \frac{1000(1.06)(1.06^{15} - 1)}{1.06 - 1} \\ &= \underline{\underline{\$24673}} \end{aligned}$$

(10)

a) i) $M_0 = 100$

and $60 = 100 e^{-35k}$

$$e^{-35k} = \frac{3}{5}$$

$$-35k = \log\left(\frac{3}{5}\right)$$

$$\underline{k = -\frac{1}{35} \log\left(\frac{3}{5}\right)}$$

ii) $\frac{dM}{dt} = -kM$
 $= \frac{1}{35} \log\left(\frac{3}{5}\right) \times 50$
 $\doteq 0.73 \text{ g/m}$

iii) $5 = 100 e^{-kt}$
 $\frac{1}{20} = e^{-kt}$
 $\therefore -kt = \log\left(\frac{1}{20}\right)$
or $kt = \log 20$
 $\therefore t = \frac{\log 20}{k}$
 $\doteq 205 \text{ min.}$

b) i) $t=0, x = 0 \times e^0$

$$\underline{x = 0}$$

ii) $v = \frac{dx}{dt}$
 $= 2 \times e^{-t} + e^{-t} \times (-1) \times 2t$
 $= 2e^{-t} - 2te^{-t}$
 $= 2e^{-t}(1-t)$

iii) $v = 0 \Rightarrow \underline{t = 1 \text{ second}}$

iv) $x = \frac{2t}{e^t}$
as $t \rightarrow \infty, \underline{x \rightarrow 0^+}$

